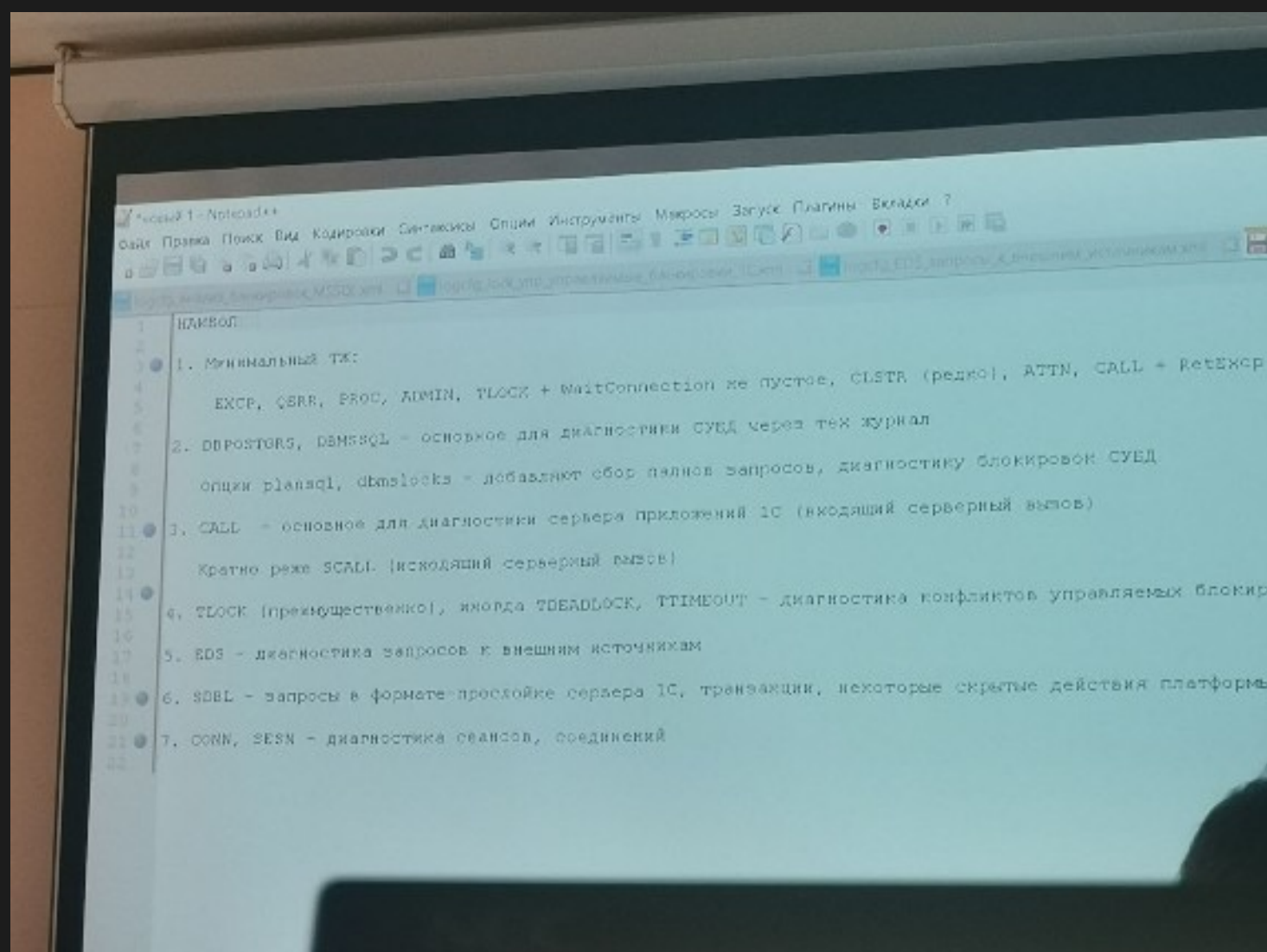
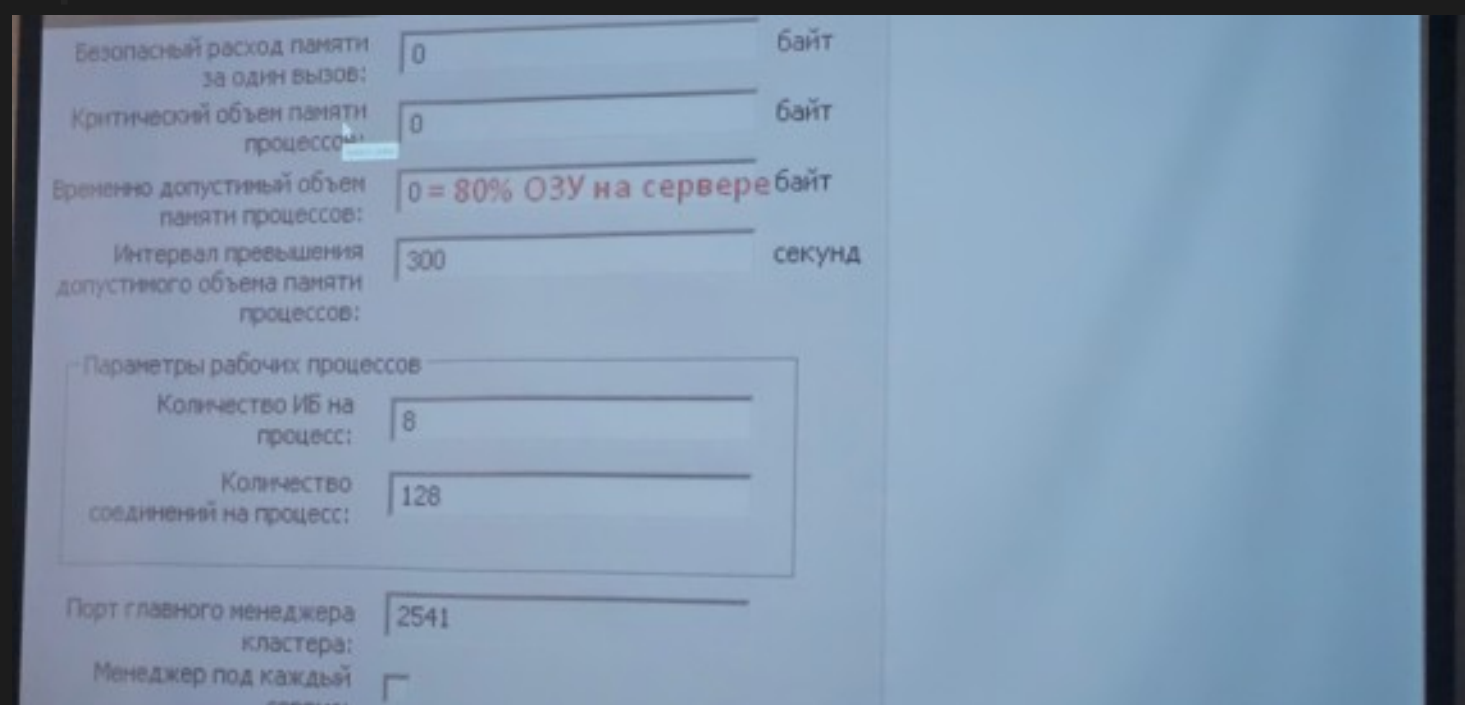


Наиболее активно использ. настройки ТЖ



Настройки кластера временное привышение памяти



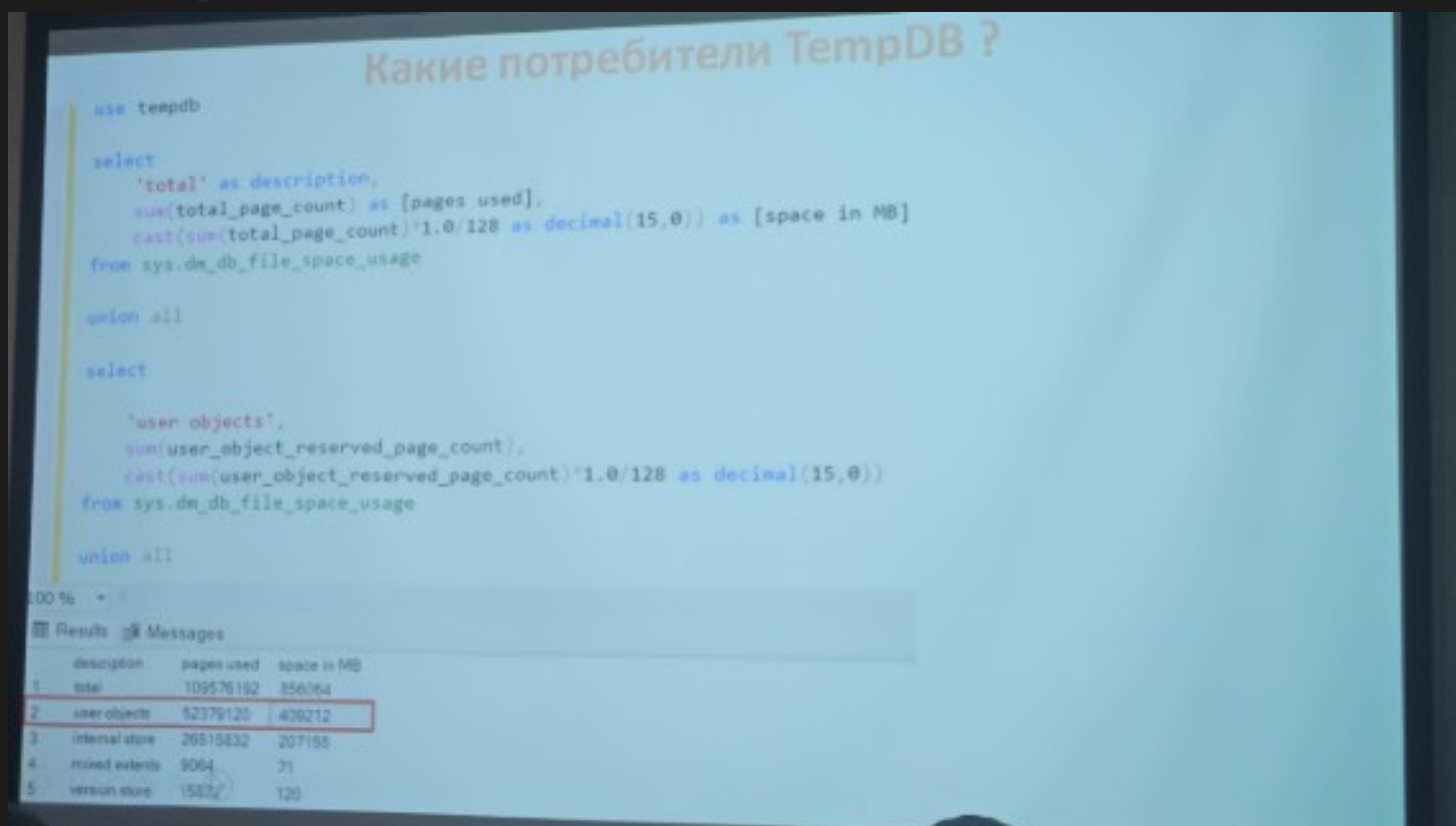
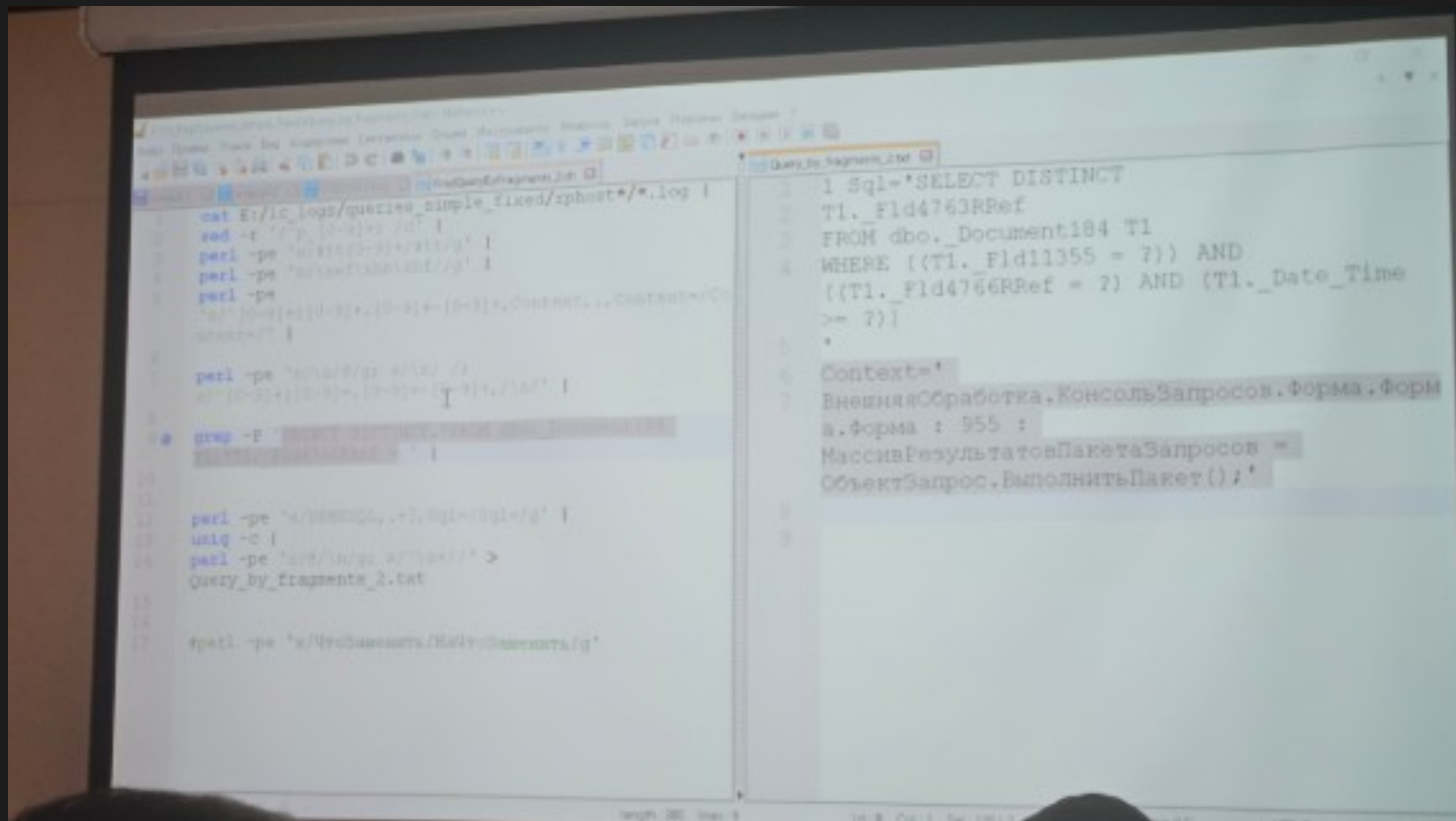
Процессор нагружен когда перекидываются сеансы между процессами

Пинг процессов 1с

```
47:33.199011-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:00.213011-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:05.354001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:10.244010-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:15.351001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:20.304001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:20.904004-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:25.320000-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:25.351021-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:30.320001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:30.413014-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:42.585001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:42.495024-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:47.585010-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:47.717014-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:47.773050-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process is abandoned, agentURL=tcp://
, procURL=tcp://
48:52.432009-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:52.710012-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:57.432010-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
48:57.945033-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
48:57.974020-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process terminated, agentURL=tcp://
, procURL=tcp://m
49:02.414011-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
49:02.443035-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
49:07.433011-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
49:07.479010-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
49:12.493007-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
49:12.038012-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
49:17.401001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
49:16.180007-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
49:22.494001-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit and would be stopped
49:23.024000-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process exceeded temporary allowed memory limit exceeded for server, ServerID=55a1dec7-
49:30.430004-0, ATTN, 1, process=agent, OSThread=6688, Descr=Process is abandoned, agentURL=tcp://
, procURL=tcp://m
49:30.430004-0, ATTN, 1, process=agent, OSThread=4988, Descr=Process terminated, agentURL=tcp://
, procURL=tcp://m
```

Настройка кластера критический объем памяти

Безопасный расход памяти за один вызов:	0	байт
Критический объем памяти процессов:	0= 95% памяти сервера	байт
Временно допустимый объем памяти процессов:	-1	байт
Интервал превышения допустимого объема памяти процессов:	300	секунд
Параметры рабочих процессов		
Количество ИБ на процесс:	8	
Количество соединений на процесс:	128	
Порт главного менеджера кластера:	2541	



```

select i10 as i0,
       sum(i1.Execs) as Execs
into #ConsumedTempdb
from
(
select
  writes = total_logical_writes,
  IO = (total_logical_reads + total_logical_writes),
  CPU = total_worker_time,
  Execs = qs.execution_count,
  sql_text = substring (qt.text, qs.statement_start_offset/2,
    (case when qs.statement_end_offset = -1
      then len(convert(nvarchar(max), qt.text)) * 2
    else qs.statement_end_offset end - qs.statement_start_offset)/2)
from
  sys.dm_exec_query_stats qs
cross apply
  sys.dm_exec_sql_text(qs.sql_handle) as qt
) i1
where
  writes > 1000
and i1.sql_text like 'INSERT INTO #ttt' or i1.sql_text like 'SELECT INTO #ttt'
or i1.sql_text not like 'INSERT'
and sql_text not like 'DELETE'
and sql_text not like 'UPDATE'
and sql_text not like 'CREATE'
and sql_text not like 'TRUNCATE'
and sql_text not like 'DROP'
group by

```



Поиск в Тж запроса джейсон
Cat grep

Индексы

Условия использования индекса

1. SARG (Searchable ARGuments): условия, позволяющие использовать индекс

```

=
<
>
<=
>=

```

BETWEEN (МЕЖДУ)

IN (В)

LIKE (ПОДОБНО) 'подстрока'

(работает быстро, если у 'подстрока' нет масок в начале, например LIKE 'Иванов%')

ORDER BY (УПОРЯДОЧИТЬ ПО)

(правильнее сказать - не аргумент поиска, а использует индекс)

2. Non SARG: условия, НЕ использующие индекс

Not Like

Not IN

Field + '123' = 'MyField'

1. **SEEK** для составного индекса:

```

ВЫБРАТЬ
*
ИЗ
    РегистрСведений.Тестовый КАК Тестовый
ГДЕ

    // <Seek predicates> или <Условие поиска>

    Тестовый.Измерение1 = @Измерение1
И Тестовый.Измерение2 = @Измерение2
И Тестовый.Измерение3 = @Измерение3

ВЫБРАТЬ
*
ИЗ
    РегистрСведений.Тестовый КАК Тестовый
ГДЕ

    // <Seek predicates> или <Условие поиска>

    Тестовый.Измерение1 = @Измерение1
И Тестовый.Измерение2 = @Измерение2
//И Тестовый.Измерение3 = @Измерение3 // пропущенные поля в конце индекса

```

2. SEEK ... WHERE для Postgres

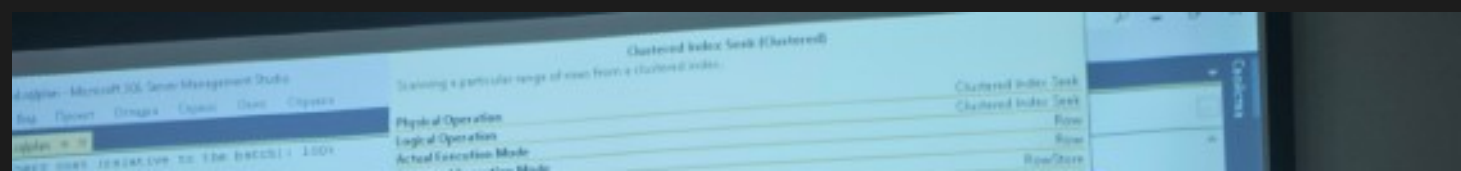
1 SELECT _idref, _number
FROM _document18
3 WHERE _number = '0000001' AND _posted = true

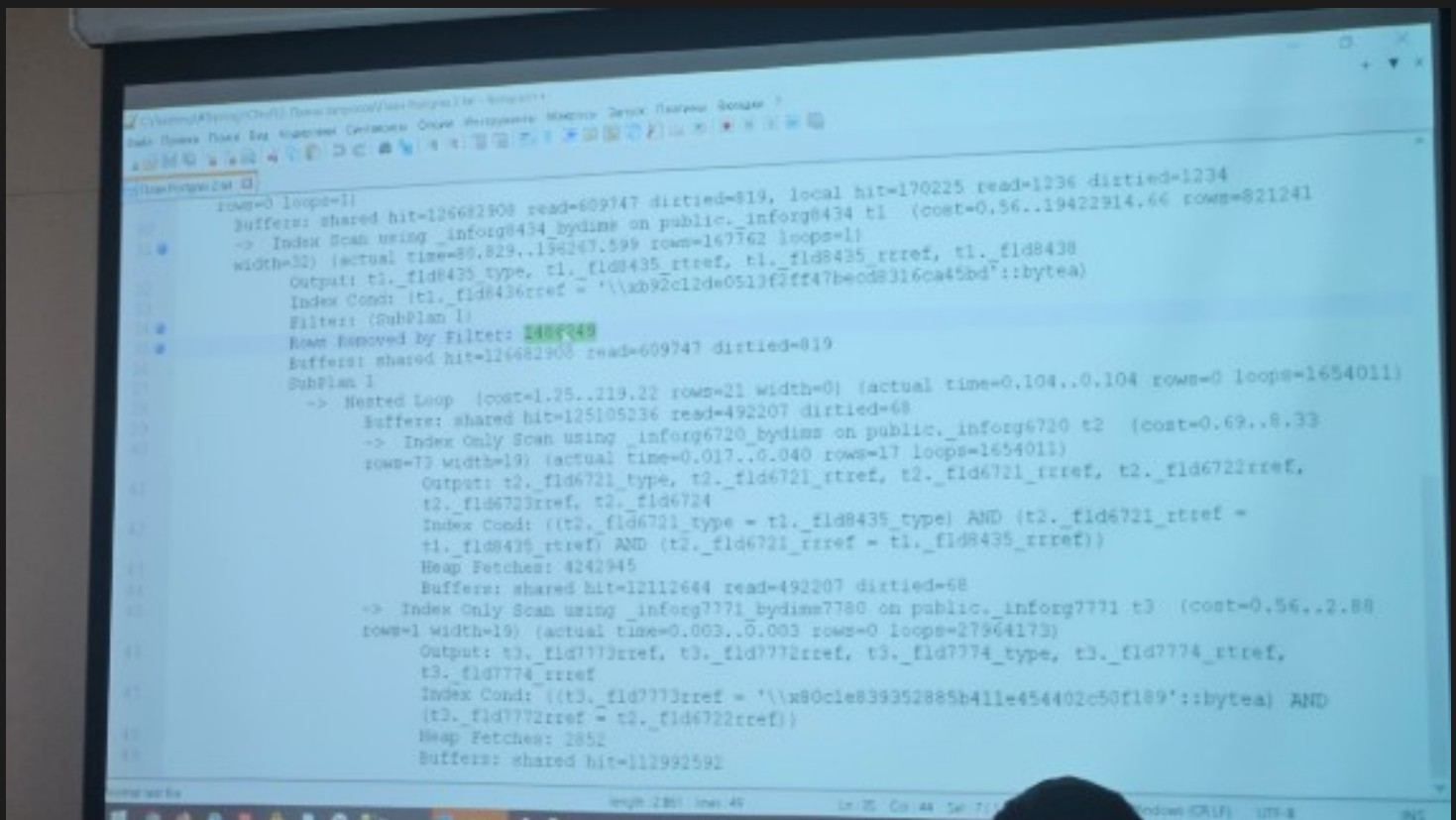
Data Output Explain Messages Notifications

Field	Type	Value
_idref	Integer	0
_number	String	0000001
_posted	Boolean	true

WHERE

WHERE _number = 0000001

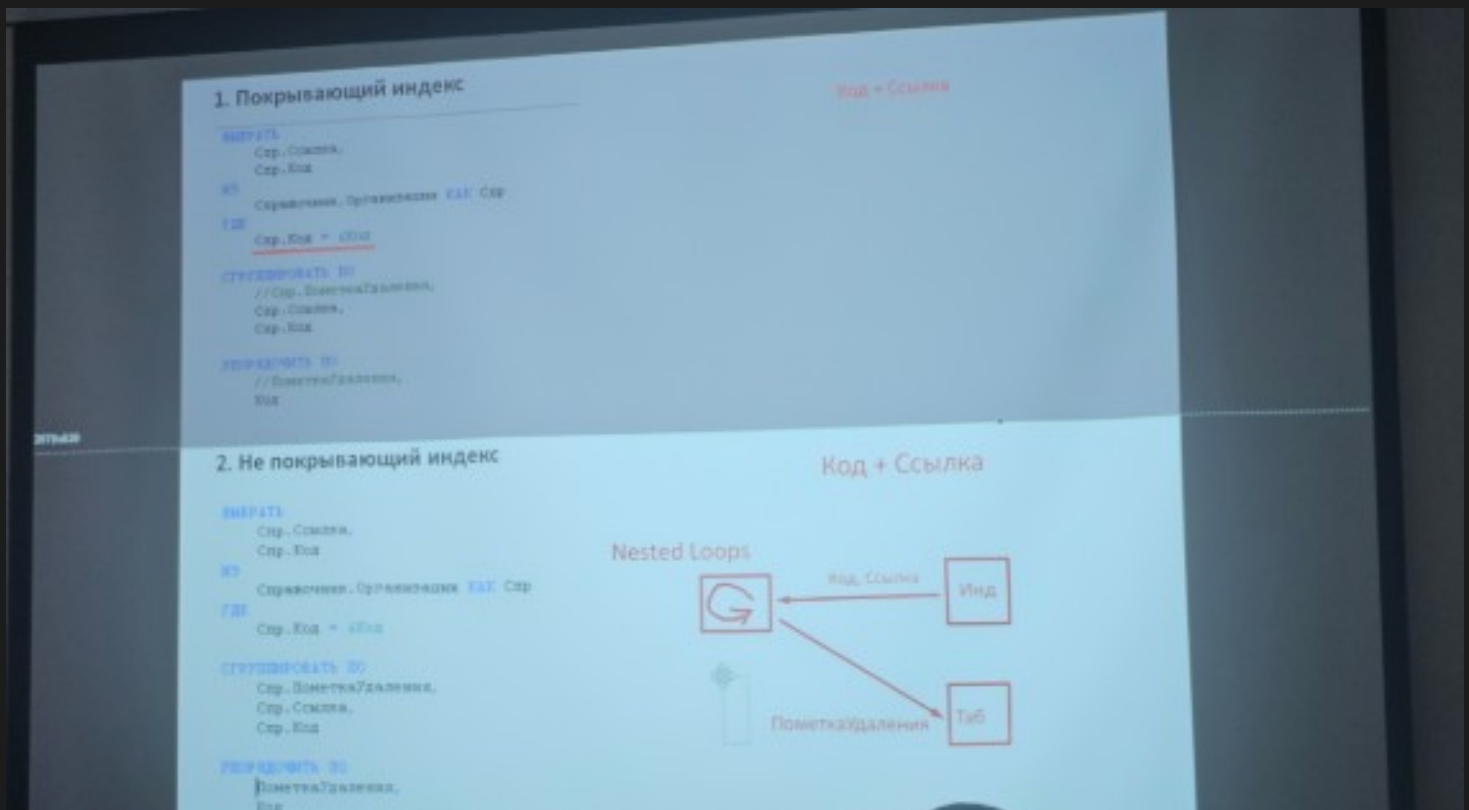




<https://1c.ru/news/info.jsp?id=24500>

Нагрузочный тест ерп

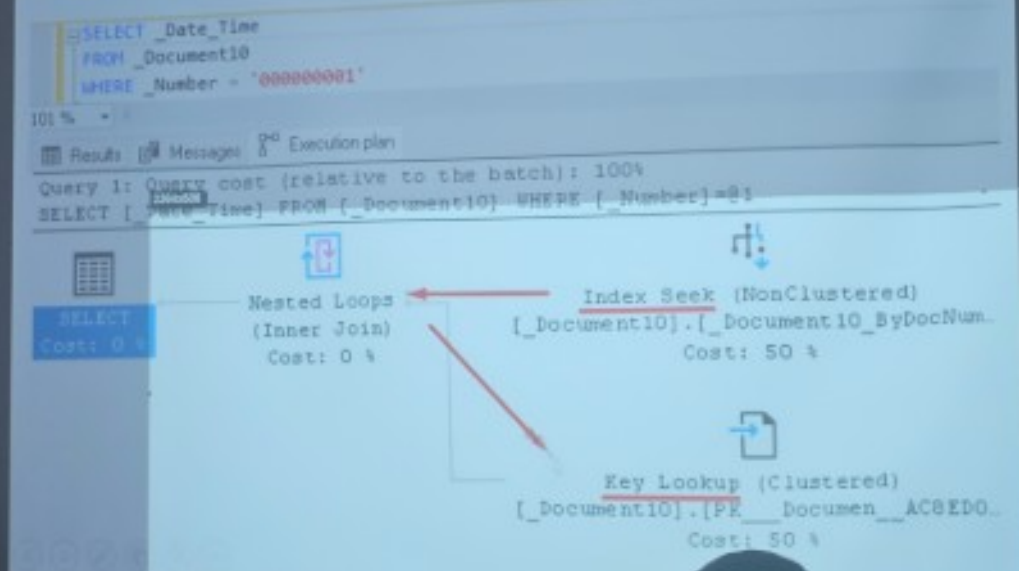
Nested loops



Key lookup/rid lookup. Не покрывающий индекс

2. Не покрывающий индекс MS SQL Server

Index seek + RID Lookup/Key lookup



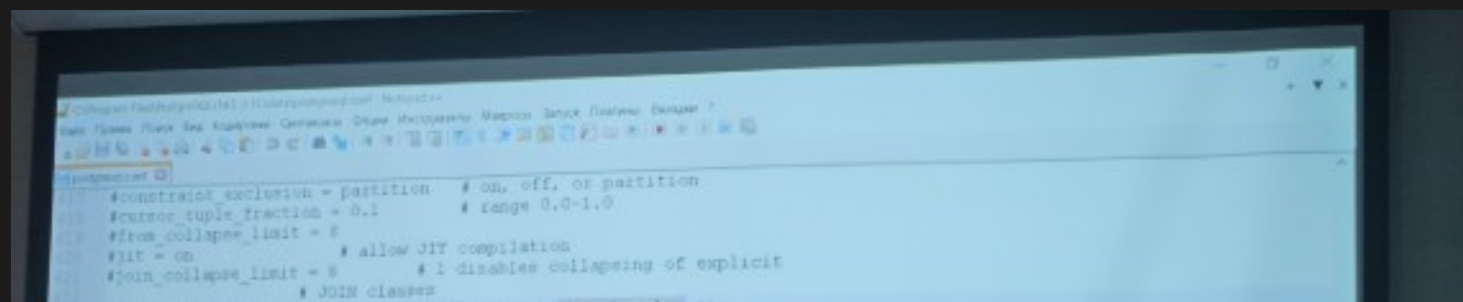
Постегрес. Настройка подбора плана запроса

Generic plan-как в Скуле до 2022.

Кешурет план

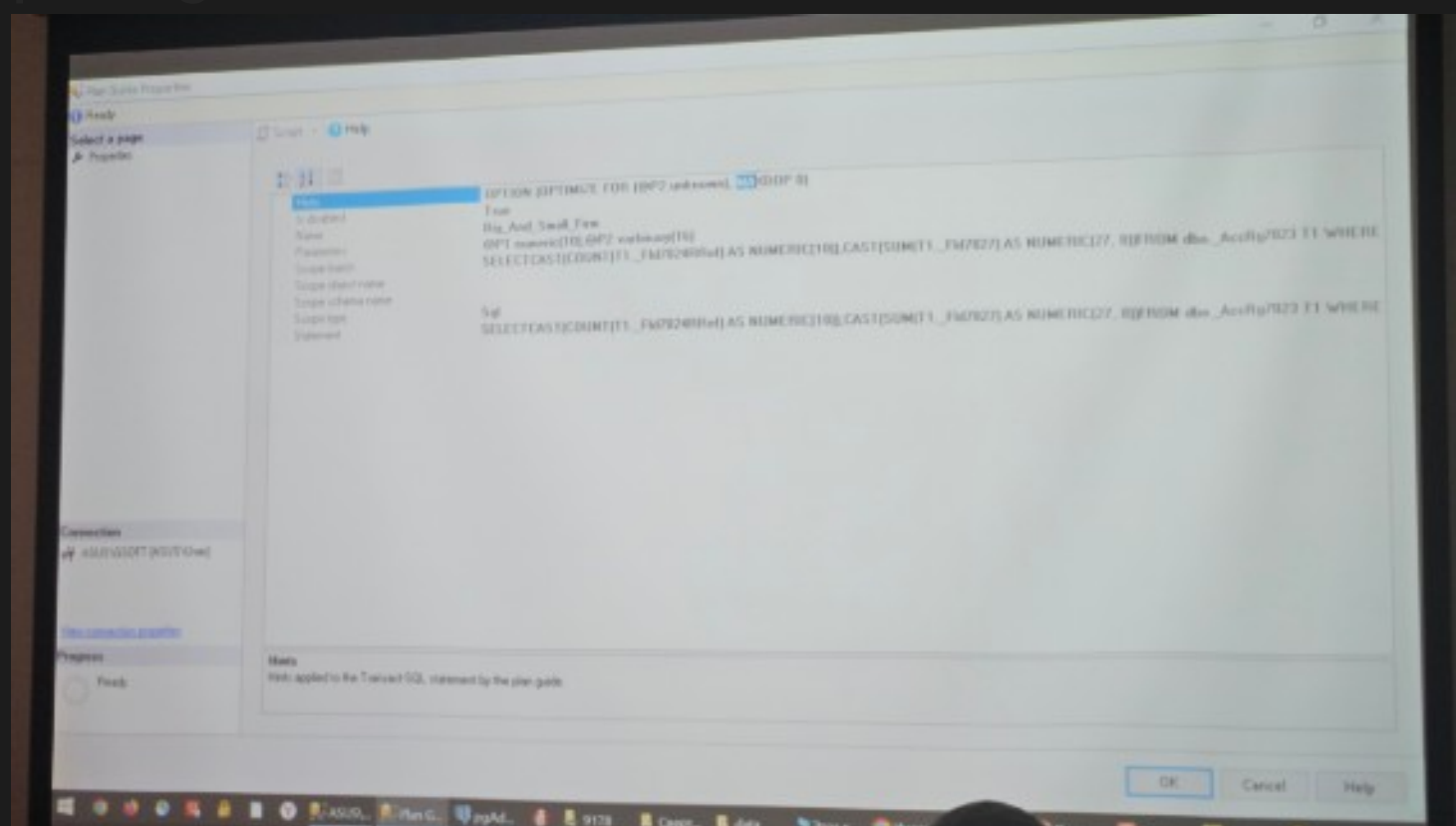
Force custom - без Кеша, всегда будет строить план

Auto-несколько раз планы подбираются




```
424 #plan_cache_mode = auto          # auto, force_generic_plan or
425 # force_custom_plan
426
427 -----
428 # REPORTING AND LOGGING
429 -----
430
431 # - Where to Log -
432
433 log_destination = 'stderr'        # Valid values are combinations of
434 # stderr, csvlog, syslog, and eventlog,
435 # depending on platform. csvlog
436 # requires logging_collector to be on.
437
438 # This is used when logging to stderr:
439 logging_collector = on            # Enable capturing of stderr and csvlog
440 # into log files. Required to be on for
441 # csvlogs.
442 # (change requires restart)
443
444 # These are only used if logging_collector is on
445 #log_destination = 'file'         # destination for log file
446 #log_filename = 'postgresql-%Y-%m-%d.log' # log file name, with %Y, %m, %d
447 # replaced by date of day in YYYY-MM-DD format
448 # The default is to log to stderr and no file.
449 # Logging to log file is dependent on logging_collector
450 # being on, and then log_destination must be set
451 # to a file.
```

Перехват запроса для изменения плана plan guide



Эффективные запросы:

<https://its.1c.ru/db/v8std#content:658:hdoc@6c55c3e5>

